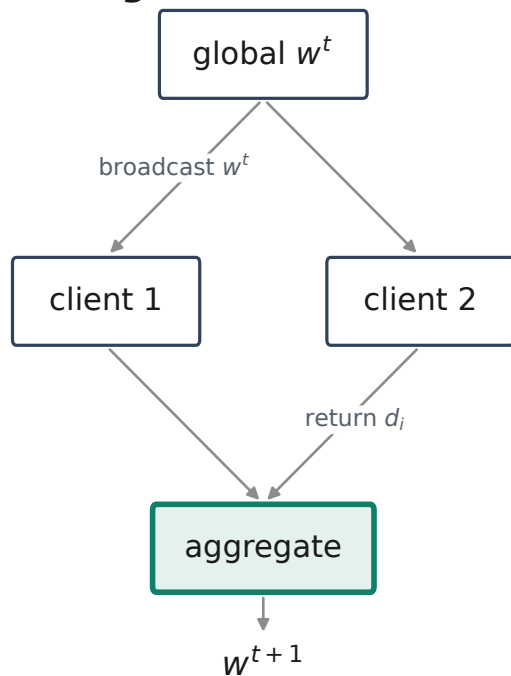
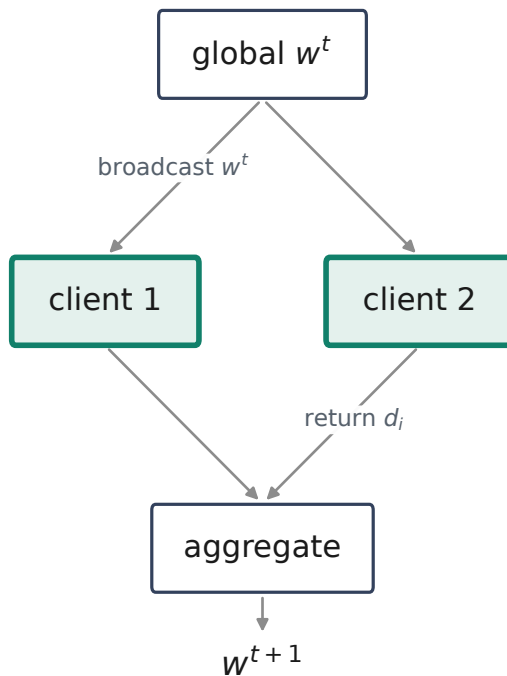


(a) FedAvg

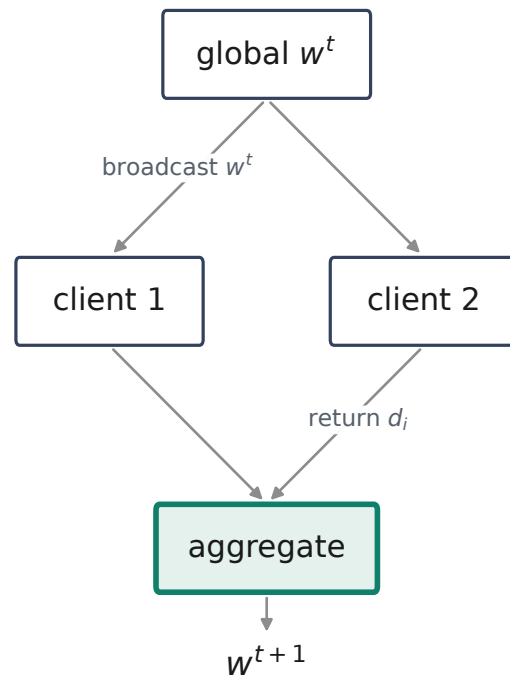
aggregation: sample-weighted

$$w^{t+1} = \sum_i \frac{n_i}{N} w_i^{t+1}$$

(b) FedProx

local objective: + proximal term

$$\mathcal{L}_{\text{CE}}(w) + \frac{\mu}{2} \|w - w^t\|^2$$

(c) FedNova

aggregation: work-normalised

$$w^{t+1} = w^t + a_{\text{eff}} \sum_i p_i \frac{d_i}{a_i}$$

$a_i = \tau_i$ (effective local work)